Early Black-Hole Growth Constraints from JWST

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Introduction

How did early black holes grow so big, so fast? The James Webb Space Telescope (JWST) has revealed massive black holes with little time to grow.[1] I compiled a catalogue of published black-hole mass estimates and compared their growth requirements, testing sensitivity to mass errors, seed mass, growth onset, and radiative efficiency.

Methods

Catalogue creation and sampling

350 measurements from 32 source families at z ≥ 4

338 provisional objects

234 expanded-sample objects

220 primary
Broad hydrogen (Balmer) line masses [2]

14 additional:
- 7 masses with uncertain line origins, assuming gas orbiting the black hole
- 6 tentative broad-line detections
- GN-z11: ultraviolet-line mass [1]

Outside the growth sample:
- 98 without an eligible mass
- 6 with uncertain identity

Symbols: M⊙ denotes solar mass. Higher redshift z means earlier times. ε is the radiated fraction of accreted rest-mass energy.

Eddington ratio: f_{Edd} = luminosity/Eddington luminosity, where radiation force balances gravity. Averages above 1 require super-Eddington episodes.[3]

Growth test and uncertainty

Standardized mass, redshift and reported errors;
Checked identity, adopted one mass per object, retained alternatives

Central mass estimate

10,000 mass-error draws per object with reported errors

Calculated the required lifetime-average Eddington ratio, including inactive periods [3]
Reference: 100 M⊙ seeds at z = 30
ε = 0.1, no mergers

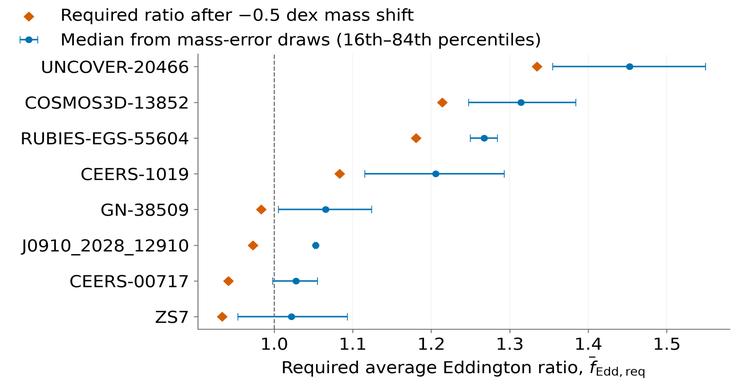
Ranked central estimates; Compared required average with 1

Calculated the fraction of mass-error draws above 1

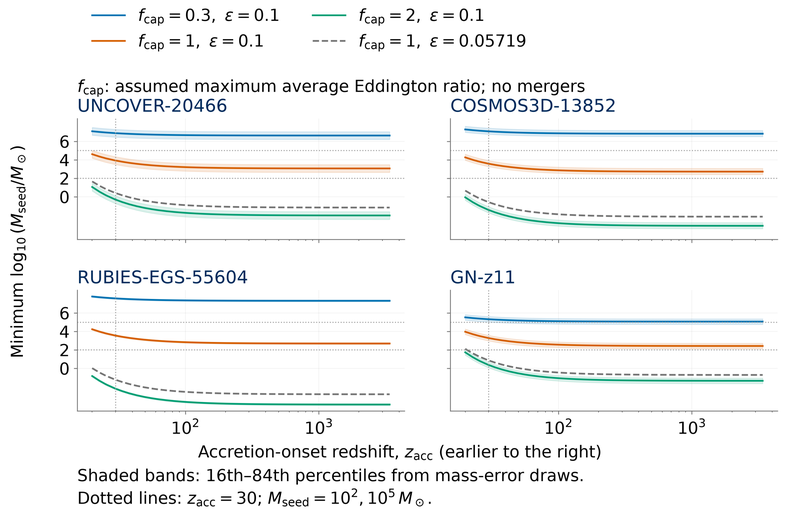
Uncertainty: Mass errors cover 208 primary / 222 expanded objects. The other 12 enter central-mass tests only. Reported mass errors do not consistently include calibration uncertainty.
Earlier onset: Solved minimum seed masses for 4 targets at start redshifts 20–3,400. Included radiation in cosmic ages and assumed gas was available.

Results

- **Reference model:** 8/220 primary objects require average Eddington ratios above 1, including 5 in ≥ 95% of mass-error draws.
- **Mass sensitivity:** Lowering masses by 0.5 dex (a factor of 3.2) leaves 3 primary objects requiring averages above 1 in ≥ 95% of draws.
- **Seed mass:** 1,000 M⊙ seeds leave 3/220 above 1. With 100,000 M⊙ seeds, neither sample has objects above 1.
- **Growth start time:** At z = 20, 18/220 primary objects require averages above 1, and GN-z11 requires the highest average in the expanded sample.
- **Efficiency:** At ε = 0.05719 (non-spinning thin disc) [4], all 234 central mass estimates require averages below 1 under the other reference assumptions.



- **Earlier-onset test:** Starting at matter–radiation equality (z = 3,400) still requires 270–1,200 M⊙ seeds for the 4 plotted objects (ε = 0.1, average ratio ≤ 1, no mergers).



Earlier starts lower the required seed mass, but the decrease becomes smaller.

- **Independent mass check:** A2744-QSO1's broad-line mass (2.0×10^7 M⊙) requires an average of 0.929 under the reference model. Its dynamical mass (5.0×10^7 M⊙)[5] requires 0.9996. Its required average spans 0.947–1.052 (16th–84th percentiles), including 1.

Conclusions

- **Catalogue:** The reproducible catalogue identifies the strongest targets for growth tests by comparing published masses under the same assumptions.
- **Growth physics:** The efficiency test shows that difficulty growing from small seeds does not imply a need for heavier seeds or super-Eddington accretion.
- **Earlier growth:** Earlier onset alone cannot explain the four target masses from 100 M⊙ seeds at ε = 0.1 and average ratio ≤ 1, without mergers.
- **Future observations:** The independent mass check shows that the inferred growth difficulty depends on the mass estimate. Independent masses and checks on lensing and line broadening would test whether the strongest outliers truly challenge the reference growth model.

References

[1] Maiolino et al. (2024), Nature, 627, 59.
[2] Shen (2013), BASI, 41, 61.
[3] Dayal (2024), A&A, 690, A182.
[4] Bardeen et al. (1972), ApJ, 178, 347.
[5] Juodzbalis et al. (2025), arXiv:2508.21748v2.

Catalogue sources and code: QR link above.